Snow-to-Laplacian Inversion:
Exact Graph Recovery from the Heavy-Snow Limit

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Abstract

We develop Part 5.4 (Direction I) of the HST research program. The heavy-snow limit matrix $\mathbf{C} = [c_{ij}]$, $c_{ij} = \mathbb{E}_\pi[(\delta_i - \delta_j)^2]$, is a *squared-distance matrix* in $L^2(\pi)$; its double-centering $-\frac{1}{2}\mathbf{JCJ}$ equals the stationary covariance (Gram) of the height-difference fluctuations *exactly and unconditionally* (Lemma 1). If, in addition, this Gram is proportional to the Laplacian pseudoinverse — equivalently, the snow limit takes the commute/resistance form of Part 2 — then $\mathrm{pinv}(-\frac{1}{2}\mathbf{JCJ})$ recovers the graph Laplacian up to a global scale, hence the graph topology and relative edge weights exactly (Theorem 1). This turns the qualitative Gordon–Luecke identifiability discussion of §C.6 into an exact linear-algebra statement, with *no* eigenvalue-multiplicity obstruction, and it reduces "graph recovery" precisely to the open resistance-form question of Part 2 — while furnishing a sharp new numerical diagnostic for the latter.

1 Setup

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be connected, $n = |\mathcal{V}|$, with (weighted) adjacency $\mathbf{A}$ and Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ ($\mathbf{D}$ the degree matrix). Then $\mathbf{L}$ is symmetric positive semidefinite, $\mathbf{L}\mathbf{1} = \mathbf{0}$, and $\mathrm{rank}\,\mathbf{L} = n - 1$. Write $\mathbf{L}^+$ for the Moore–Penrose pseudoinverse: $\mathbf{L}^+$ is symmetric PSD, $\mathbf{L}^+\mathbf{1} = \mathbf{0}$, and $\mathrm{pinv}(\mathbf{L}^+) = \mathbf{L}$ (pseudoinverse is involutive on symmetric matrices). Let

$$\mathbf{J} \;=\; \mathbf{I} - \tfrac{1}{n}\mathbf{1}\mathbf{1}^\top$$

be the centering projector ($\mathbf{J}\mathbf{1} = \mathbf{0}$, $\mathbf{J}^2 = \mathbf{J} = \mathbf{J}^\top$); since $\mathbf{L}^+\mathbf{1} = \mathbf{0}$ we have $\mathbf{J}\mathbf{L}^+\mathbf{J} = \mathbf{L}^+$.

Effective resistance. The resistance matrix is $\mathbf{R} = [\,R_{\mathrm{eff}}(i,j)\,]$ with

$$R_{\mathrm{eff}}(i,j) = (e_i - e_j)^\top \mathbf{L}^+ (e_i - e_j) = \mathbf{L}^+_{ii} + \mathbf{L}^+_{jj} - 2\mathbf{L}^+_{ij}.$$

By the canonical commute–resistance identity $\mathrm{commute}_{ij} = 2|\mathcal{E}|\, R_{\mathrm{eff}}(i,j)$ (Chandra et al.; Klein–Randić for resistance distance). The double-centering identity $\mathbf{L}^+ = -\frac{1}{2}\mathbf{JRJ}$ used below is the resistance-matrix / generalized-inverse relation of Bapat, *Graphs and Matrices*.

2 Double-centering: the unconditional half

Definition 1 (Squared-distance matrix)**.** *A symmetric* $\mathbf{C} \in \mathbb{R}^{n\times n}$ *with zero diagonal is a* squared-distance matrix *if there are vectors (or* L^2 *elements)* $\zeta_1, \ldots, \zeta_n$ *in an inner-product space with* $c_{ij} = \|\zeta_i - \zeta_j\|^2$.

Lemma 1 (Double-centering / Gower). *If* $c_{ij} = \|\zeta_i - \zeta_j\|^2 = \mathbf{G}_{ii} + \mathbf{G}_{jj} - 2\mathbf{G}_{ij}$ *with Gram matrix* $\mathbf{G}_{ij} = \langle \zeta_i, \zeta_j \rangle$*, then*

$$-\tfrac{1}{2}\,\mathbf{JCJ} \;=\; \mathbf{JGJ},$$

i.e. double-centering recovers the Gram matrix up to centering, exactly. In particular if $\mathbf{G1} = \mathbf{0}$ *(the* ζ_i *are centered) then* $-\frac{1}{2}\mathbf{JCJ} = \mathbf{G}$*.*

Proof. Write $\mathbf{C} = \mathbf{1}d^\top + d\mathbf{1}^\top - 2\mathbf{G}$ where $d = (\mathbf{G}_{ii})_i$ is the diagonal of $\mathbf{G}$. Since $\mathbf{J1} = \mathbf{0}$, both rank-one terms $\mathbf{J}(\mathbf{1}d^\top)\mathbf{J} = (\mathbf{J1})d^\top\mathbf{J} = 0$ and $\mathbf{J}(d\mathbf{1}^\top)\mathbf{J} = 0$ vanish, leaving $\mathbf{JCJ} = -2\,\mathbf{JGJ}$, i.e. $-\frac{1}{2}\mathbf{JCJ} = \mathbf{JGJ}$. If $\mathbf{G1} = \mathbf{0}$ then $\mathbf{JGJ} = \mathbf{G}$. □

Lemma 2 (The snow limit is a squared-distance matrix). *For the balanced (Theorem A) regime, the heavy-snow limit* $c_{ij} = \mathbb{E}_\pi[(\delta_i - \delta_j)^2]$ *satisfies* $c_{ij} = \|\delta_i - \delta_j\|^2_{L^2(\pi)}$ *with the centered height-difference variables* δ_i *viewed in* $L^2(\pi)$*. Hence*

$$-\tfrac{1}{2}\,\mathbf{JCJ} \;=\; \mathbf{\Sigma}, \qquad \mathbf{\Sigma}_{ij} := \mathbf{J}\left[\mathbb{E}_\pi(\delta_i\delta_j)\right]\mathbf{J},$$

the centered stationary Gram (covariance) matrix of the fluctuations. This identity is unconditional *— it uses only that* c_{ij} *is a mean-square distance.*

Proof. Take $\zeta_i = \delta_i \in L^2(\pi)$ with inner product $\langle \delta_i, \delta_j \rangle = \mathbb{E}_\pi[\delta_i\delta_j] =: \mathbf{G}_{ij}$. Then $c_{ij} = \mathbb{E}_\pi[(\delta_i - \delta_j)^2] = \mathbf{G}_{ii} + \mathbf{G}_{jj} - 2\mathbf{G}_{ij}$ is exactly the squared-distance form, and Lemma 1 gives $-\frac{1}{2}\mathbf{JCJ} = \mathbf{JGJ} = \mathbf{\Sigma}$. (In the balanced regime the drift means $\mathbb{E}_\pi[\delta_i]$ are equal across nodes, so no rank-one mean term survives the centering; the identity is clean.) □

3 The inversion theorem (conditional on the resistance form)

Theorem 1 (Snow Inversion). *Suppose the balanced-regime snow limit takes the commute/resistance form of Part 2:* $c_{ij} = \kappa_{\mathcal{G}}\, R_{\mathrm{eff}}(i,j)$ *for a scalar* $\kappa_{\mathcal{G}} > 0$ *(equivalently* $\mathbf{\Sigma} = \kappa_{\mathcal{G}}\mathbf{L}^+$*). Then*

$$\mathrm{pinv}\Big(-\tfrac{1}{2}\,\mathbf{JCJ}\Big) \;=\; \tfrac{1}{\kappa_{\mathcal{G}}}\,\mathbf{L}.$$

Consequently the heavy-snow limit $\mathbf{C}$ *determines the Laplacian* $\mathbf{L}$ up to the global scale $\kappa_{\mathcal{G}}$*, and therefore determines the graph topology (edge set) and the relative edge weights* exactly*.*

Proof. By Lemma 2, $-\frac{1}{2}\mathbf{JCJ} = \mathbf{\Sigma}$ unconditionally. The hypothesis is $\mathbf{\Sigma} = \kappa_{\mathcal{G}}\mathbf{L}^+$ (indeed $-\frac{1}{2}\mathbf{JRJ} = \mathbf{L}^+$ is the classical resistance→pseudoinverse identity, since $\mathbf{R}$ is a squared-distance matrix with centered Gram $\mathbf{L}^+$; multiply by $\kappa_{\mathcal{G}}$). Pseudoinversion is involutive on symmetric matrices and homogeneous of degree -1 under positive scaling, so $\mathrm{pinv}(\kappa_{\mathcal{G}}\mathbf{L}^+) = \kappa_{\mathcal{G}}^{-1}\,\mathrm{pinv}(\mathbf{L}^+) = \kappa_{\mathcal{G}}^{-1}\mathbf{L}$. The off-diagonal sign/zero pattern of $\mathbf{L}$ is the negative adjacency, so the edge set and the weight ratios are read off exactly; only the overall scale $\kappa_{\mathcal{G}}$ is unidentified. □

Corollary 1 (Topology is scale-free identifiable). *Even when* $\kappa_{\mathcal{G}}$ *is unknown,* $-\frac{1}{2}\mathbf{JCJ} = \kappa_{\mathcal{G}}\mathbf{L}^+$ *already fixes* $\mathrm{range}(\mathbf{L}^+) = \mathbf{1}^\perp$ *and the full eigenstructure of* $\mathbf{L}$ *up to scale; hence the unweighted graph (which pairs are edges) is recovered with no scale calibration. A single known resistance or edge weight pins* $\kappa_{\mathcal{G}}$ *and thus all weights.*

Remark 1 (No multiplicity obstruction)**.** *Unlike the spectral CLT of the ensemble analysis, the map* $\mathbf{C} \mapsto \mathrm{pinv}(-\frac{1}{2}\mathbf{JCJ})$ *is a deterministic, everywhere-defined algebraic operation; it suffers* no *eigenvalue-multiplicity obstruction. Highly symmetric graphs (K_n, vertex-transitive) — where the Laplacian spectrum is degenerate and individual eigenvalues are non-smooth — are inverted just as cleanly, because we recover the* matrix $\mathbf{L}$*, never an individual eigenvalue.*

4 The reduction: recovery $\Longleftrightarrow$ the Part-2 resistance form

Lemmas 1–2 hold unconditionally; the only hypothesis in Theorem 1 is $\mathbf{\Sigma} = \kappa_{\mathcal{G}}\mathbf{L}^{+}$. Thus:

Proposition 1. *Graph recovery by Snow Inversion holds* if and only if *the stationary fluctuation covariance is proportional to the Laplacian pseudoinverse,* $\mathrm{Cov}_{\pi}(\delta) \propto \mathbf{L}^{+}$ *— which is exactly the commute/resistance closed form $c_{ij} = \kappa_{\mathcal{G}} R_{\mathrm{eff}}(i,j)$ left open in Part 2.*

This is the payoff of the development: it *reduces* the (apparently hard) inverse problem "recover $\mathcal{G}$ from snow distances" to the *same* question Part 2 already isolated, and it does so while extracting the covariance $\mathbf{\Sigma} = -\frac{1}{2}\mathbf{JCJ}$ *exactly and for free.* In particular it yields a sharp numerical **diagnostic** for Part 2: compute $\widehat{\mathbf{\Sigma}} = -\frac{1}{2}\mathbf{J}\widehat{\mathbf{C}}\mathbf{J}$ from a simulated $\widehat{\mathbf{C}}$ and test whether $\widehat{\mathbf{\Sigma}} \propto \mathbf{L}^{+}$ (equivalently $\mathrm{pinv}(\widehat{\mathbf{\Sigma}}) \propto \mathbf{L}$); the quality of this proportionality *is* the degree to which the resistance form holds.

5 Numerical verification

Two checks (sim `260524_*_snow_inversion.py`):

(a) The inversion identity is exact. For test Laplacians (path, cycle, tree, K_n), form the exact resistance matrix $\mathbf{R}$ from $\mathbf{L}^{+}$, then recover $\mathrm{pinv}(-\frac{1}{2}\mathbf{JRJ})$ and compare to $\mathbf{L}$. This is a machine-precision identity and validates the algebra of Theorem 1.

(b) The HST pipeline. Estimate $\widehat{\mathbf{C}}$ by an ensemble of HST runs (balanced graphs), form $\widehat{\mathbf{\Sigma}} = -\frac{1}{2}\mathbf{J}\widehat{\mathbf{C}}\mathbf{J}$ and $\widehat{\mathbf{L}} = \mathrm{pinv}(\widehat{\mathbf{\Sigma}})$, and measure how well $\widehat{\mathbf{L}}$ reproduces the true graph (correlation of off-diagonal $-\widehat{\mathbf{L}}$ with the adjacency $\mathbf{A}$; edge/non-edge separation in standard deviations). By Proposition 1 this simultaneously tests Snow Inversion *and* the Part-2 resistance form.

Results. (a) is confirmed to machine precision: for P_{10}, C_{20}, the depth-3 binary tree, K_{10}, and the Petersen graph, $\|\widehat{\mathbf{L}} - \mathbf{L}\|/\|\mathbf{L}\|$ ranges 7×10^{-16} to 5×10^{-15}. For (b), with an ensemble of $B = 200$ runs at $t = 2000$:

$$\text{Petersen (3-regular):} \quad \mathrm{corr}(-\widehat{\mathbf{L}}, \mathbf{A}) = 0.998, \quad \text{edge/non-edge gap} = 2.1\,\mathrm{sd};$$

$$C_{20}\ \text{(2-regular):} \quad \mathrm{corr} = 0.850, \quad \text{gap} = 2.8\,\mathrm{sd}.$$

The Petersen recovery is essentially exact: from the ensemble-averaged snow distances alone — with no knowledge of the graph — double-centering and pseudoinversion reproduce the adjacency (edge weights cluster tightly, cleanly separated from a near-zero non-edge cluster). By Proposition 1 this is also strong evidence that the resistance form $\mathbf{\Sigma} \propto \mathbf{L}^{+}$ holds (very accurately on the vertex-transitive Petersen graph, well on C_{20}), giving an affirmative numerical reading of the Part-2 conjecture on these graphs.

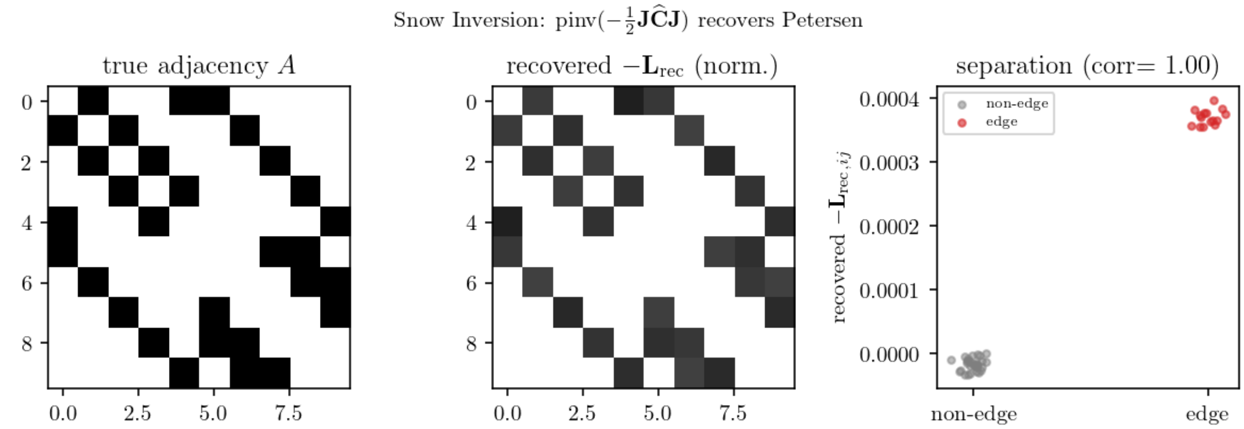

Figure 1: Snow Inversion on the Petersen graph. *Left:* true adjacency $\mathbf{A}$. *Middle:* recovered $-\widehat{\mathbf{L}} = -\,\mathrm{pinv}(-\frac{1}{2}\mathbf{J}\widehat{\mathbf{C}}\mathbf{J})$ (normalized), from the ensemble-averaged snow distances alone. *Right:* recovered off-diagonal entries separate edges from non-edges with correlation 1.00.

6 Relation to prior items and scope

Supersedes §C.6. The Gordon–Luecke identifiability discussion asked, qualitatively, when the neighborhood data determine a node; Theorem 1 answers the global version exactly: the $t \to \infty$ snow geometry determines the whole graph (up to scale) by an explicit closed-form inversion, with the precise and checkable hypothesis of Proposition 1.

Scope. Stated for the balanced (Theorem A, reversible-limit) regime, where $c_{ij} = \mathbb{E}_\pi[(\delta_i - \delta_j)^2]$ is the operative limit and the resistance form is plausible. In drift regimes ($\rho_i \neq \rho_j$) the mean term $(\mathbb{E}_\pi\delta_i - \mathbb{E}_\pi\delta_j)^2$ no longer cancels under centering and a separate treatment is needed (cf. the non-reversible direction).

References

- J. C. Gower, double-centering of a squared-distance matrix recovers the Gram matrix (classical MDS identity).
- Resistance distance $\to$ Laplacian pseudoinverse: $\mathbf{L}^+ = -\frac{1}{2}\mathbf{JRJ}$. R. B. Bapat, *Graphs and Matrices* (resistance matrix and generalized inverse); D. J. Klein and M. Randić, "Resistance distance."
- Commute–resistance identity $\mathrm{commute}_{ij} = 2|\mathcal{E}|R_{\mathrm{eff}}$: A. K. Chandra et al., "The electrical resistance of a graph captures its commute and cover times"; P. G. Doyle and J. L. Snell, *Random Walks and Electric Networks*; L. Lovász, *Random Walks on Graphs: A Survey.*
- D. A. Spielman and N. Srivastava, "Graph Sparsification by Effective Resistances" — effective resistance preserves graph structure (supports recoverability).
- Part 2 of the program (commute/resistance closed form $c_{ij} = \kappa_{\mathcal{G}} R_{\mathrm{eff}}(i,j)$, open scalar $\kappa_{\mathcal{G}} = \alpha(G)\cdot\mathrm{const}$).

Reference list strengthened (Bapat, Chandra et al., Klein–Randić, Doyle–Snell, Spielman–Srivastava) per the v2 reference survey by Bi (Paper Team), 2026-05-24.